

Vinayak Mishra

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[GitHub \(Vinayak2005917\)](#) | [LinkedIn \(vinayakmishra-cse\)](#) | [Codewars : 75+ problems solved](#)

AI & ML: PyTorch, TensorFlow, Scikit-learn, Transformers, Hugging Face, Pandas, NumPy

Agentic AI & RAG: LangChain, LangGraph, Qdrant, OpenAI API, Ollama, Embeddings, Vector Databases

Web & Visualization: Django, Flask, FastAPI, Streamlit, WebSockets, REST APIs, Supabase

AI Research Intern (The Bean Code) – Sept 2025 to Jan 2026: Developed **Nuisance-Aware Representation Refinement**, a lightweight plug-in module improving cross-dataset **DFDC AUC from 0.655 to 0.705 (+7.7%)** with only **1.9% additional overhead**. Built and evaluated on **3 datasets (FF++, DFDC, Celeb-DF)** and **5 JPEG compression levels**. [View Paper](#)

Development Intern (Shopto) – July 2025 to Oct 2025: Contributed **30+ PRs** and engineered an **Agentic Firebase Retrieval system from scratch**, along with PDF-based file generation.

Python Trainee Intern (BroskiesHub) – Jul 2025 – Aug 2025: Built 5 deployable Python projects and ranked in the **top 3 performers** out of 50 coworkers.

Ask The Right Questions: Interactive RAG Detective Game [GitHub Repo](#) | [Live Demo](#)

Built an interactive detective game that teaches RAG. Implemented Top-K semantic memory retrieval from a vector database and a UMAP-based 2D visualization of the memory space. **Optimized agentic RAG latency by ~57% (5.8s to 2.5s)** using parallel speculative retrieval, caching results while the LLM processed tool selection. **Tech Stack:** LangChain, Qdrant, FastAPI, PixiJS

Andromeda: Generative UI Based Research Agent [GitHub Repo](#) | [Live Demo](#)

Built and deployed an AI Agent that **generates interactive UIs** for learning, with tools like **web search**, **asynchronous** multi-page content extraction reducing research latency from **~89s to ~12.5s (~86%)**, and **image retrieval**. **Tech Stack:** LangChain, LangGraph, FastAPI, WebSockets, SERP API, Vercel, Render

CODORA (2nd Prize, Digitize 2025, AIT Chikmagalur) [GitHub Repo](#) | [Live Demo](#)

Led a **4-member team** to develop a **real-time collaborative workspace** with AI-assisted coding, securing **2nd place out of 150+ teams** at Digitize 2025. Engineered a scalable **asynchronous architecture** using WebSockets for multi-user document editing and integrated a Copilot-style assistant. **Tech Stack:** Django REST, WebSockets, OpenRouter API, Vercel, Render.

AMC Engineering College – B.E Computer Science Engineering

Aug 2023 – Aug 2027, Bengaluru